

Miami Property Crime and Florida Storm Activity Analysis

Exploratory analysis of 2024 monthly crime and storm patterns

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Pearson correlation	Statistical significance	Decision
r = 0.47	p = 0.12	Not statistically significant

Executive Summary

Business question. Do periods of increased storm activity correspond with higher levels of property crime, and could the pattern support future law-enforcement planning?

The analysis compared monthly Miami property-crime counts with statewide Florida storm-event frequency during calendar year 2024. The workflow combined MySQL extraction, targeted field selection, analyst-defined crime classification, duplicate and date validation, monthly aggregation, exploratory visualization, and Pearson correlation analysis.

The monthly series showed a moderate positive correlation ($r = 0.47$), but the relationship was not statistically significant ($p = 0.12$). The available evidence therefore does not support concluding that higher monthly storm-event frequency was reliably associated with increased property crime. The project is most useful as an example of end-to-end analytical workflow, judgment under imperfect data conditions, and careful interpretation of an apparent visual pattern.

Data Sources and Analytical Scope

Data sources. The original analysis used the course-provided MPD Crime Data 2024 and Storm Events 2024 datasets stored in MySQL tables `mpdcrimedata` and `stormevents2024`.

- Crime scope: Miami crime records occurring during calendar year 2024, using `CFSDate` and `IncidentType` as core analytical fields.
- Storm scope: Florida storm events occurring during 2024, using `STATE`, `EVENT_TYPE`, `BEGIN_DATE_TIME`, and `END_DATE_TIME`.
- Common unit of analysis: monthly counts for both crime and storm activity.
- Excluded fields: variables unrelated to the research question, including injury, death, crop-damage, property-damage, and narrative fields from the storm dataset.

Key Analytical Decisions

Defining Property Crime

The crime dataset did not contain a predefined field identifying which incidents should be classified as property crimes. The original workflow reviewed IncidentType values and selected incidents containing burglary, larceny, stolen vehicle, vandalism, or arson. This classification was necessary to answer the research question, but it also introduced analyst judgment that should be documented for reproducibility.

Geographic Scope

Miami crime activity was compared with statewide Florida storm-event frequency. The storm dataset did not provide a direct city field suitable for a simple Miami match. Although the source included location, zone, and coordinate fields, more precise geographic matching would require additional spatial processing or GIS methods beyond the scope of the original analysis. Statewide storm-event frequency was therefore treated as an exploratory measure of broader storm activity.

Common Time Scale

Both datasets were aggregated to calendar month so that the two sources could be compared using a consistent unit. This transformed thousands of source records into a 12-row monthly comparison dataset for exploratory visualization and correlation analysis.

Analytical Workflow

- 1. Extract** Loaded the full crime and storm-event tables from MySQL into pandas DataFrames.
- 2. Select** Retained fields relevant to incident type, dates, storm type, state, and geographic context.
- 3. Filter** Created the analyst-defined property-crime subset and filtered storm records to Florida.
- 4. Validate** Checked missing values, duplicates, date ranges, and monthly distributions.
- 5. Aggregate** Converted date fields and calculated monthly crime and storm-event counts.
- 6. Visualize** Used boxplots and monthly charts to examine variability, potential outliers, and apparent co-movement.
- 7. Test** Applied Pearson correlation to measure the strength and statistical significance of the monthly linear relationship.
- 8. Interpret** Separated visual association from statistically supported evidence and documented limitations.

Data Validation

Validation was performed before the final monthly comparison to reduce the risk of inflated counts or records falling outside the intended period. The original notebook documented the following checks:

- Selected Florida storm fields contained no missing values.
- The initial Florida storm subset contained 1,465 records; 386 duplicate rows were removed, leaving 1,079 records.

- The property-crime subset contained 16,617 records and no duplicate rows were identified in that subset.
- Property-crime dates covered January 1 through December 31, 2024.
- Florida storm-event start dates covered January 6 through December 29, 2024.
- Monthly distributions were reviewed for unusual values before correlation analysis.

Visual Analysis

The monthly plots were used first as exploratory tools. Property crime remained relatively stable across the year, while storm-event frequency varied much more sharply. The side-by-side views below show the difference in variability before the two series are compared directly.

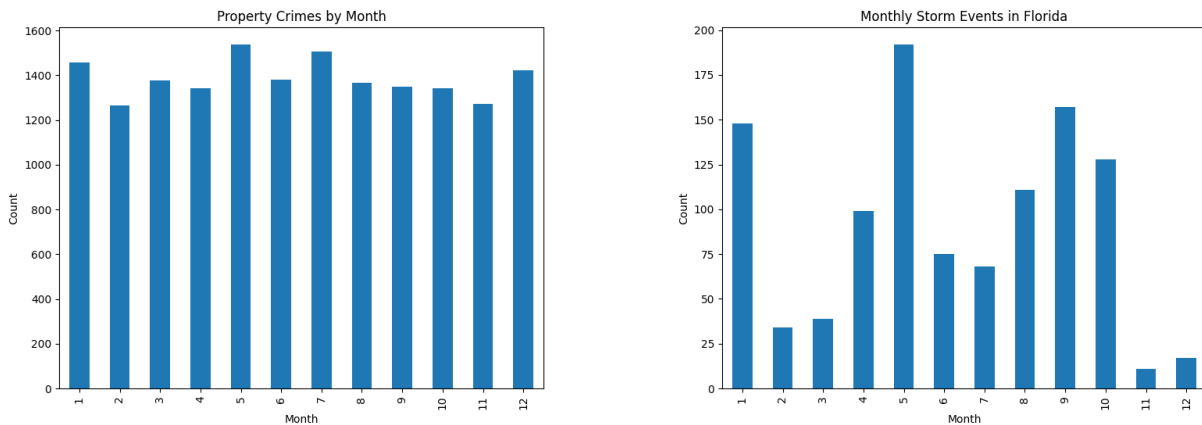


Figure 1. Monthly Miami property-crime counts and statewide Florida storm-event counts.

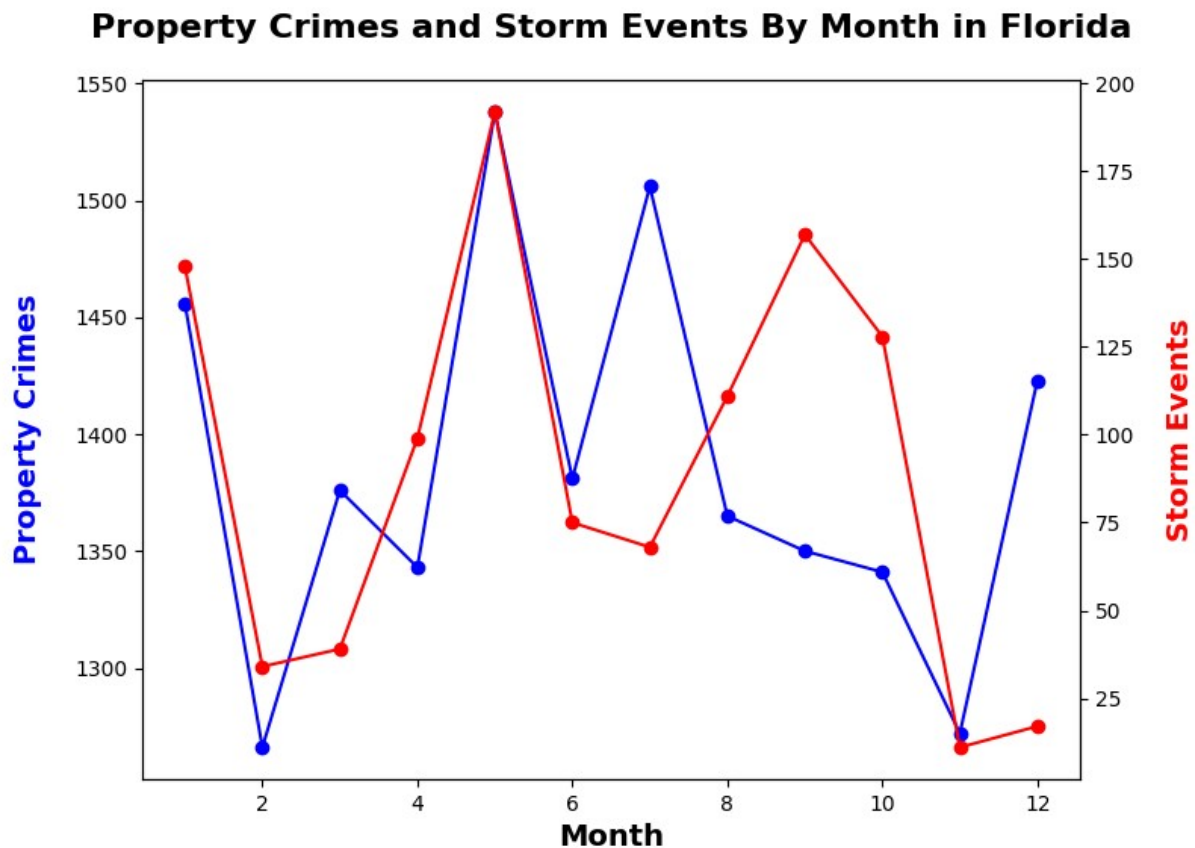


Figure 2. Original dual-axis monthly comparison. Crime and storm activity sometimes moved together, but the pattern was not consistent across all months.

Correlation Analysis

Pearson correlation was used to quantify the linear relationship suggested by the monthly comparison. The result was $r = 0.47$, indicating a moderate positive linear relationship. The associated **p-value = 0.12** exceeded the 0.05 significance level. The null hypothesis was therefore not rejected.

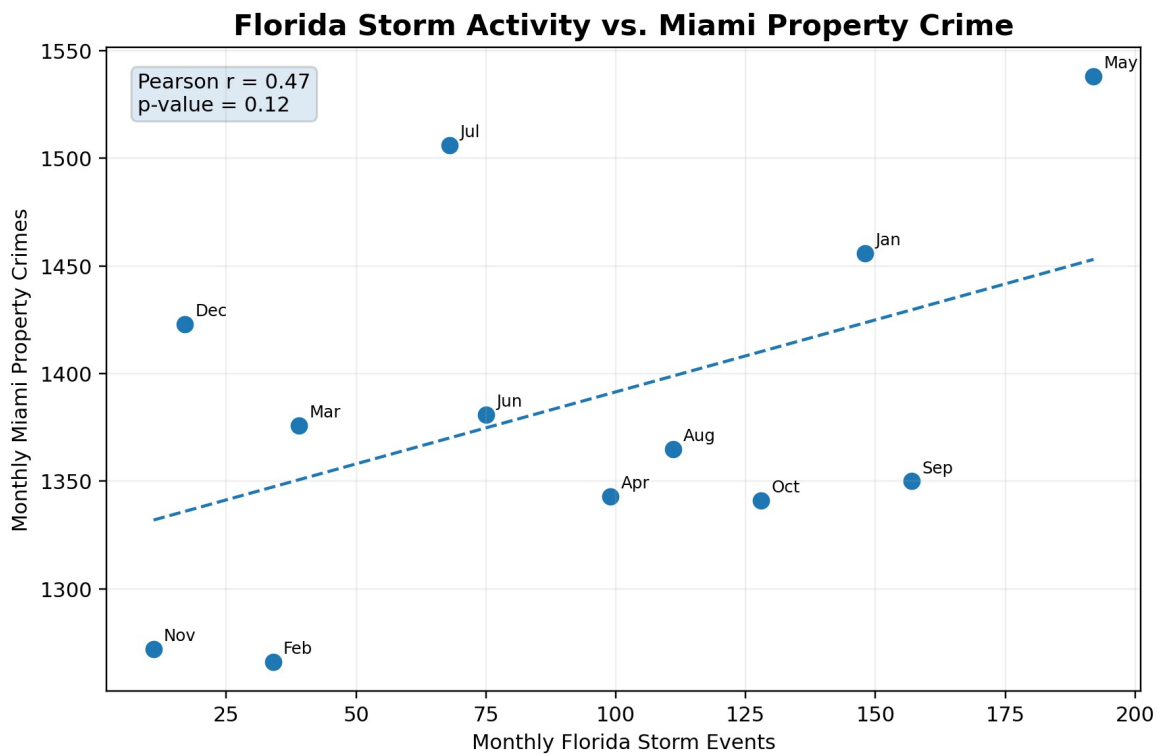


Figure 3. Scatterplot of monthly Florida storm-event frequency and Miami property-crime counts with fitted linear trend.

Findings and Interpretation

- Monthly property crime ranged from approximately 1,270 to 1,540 incidents and was comparatively stable across the year.
- Storm-event frequency showed greater month-to-month variability, with notable increases in January, May, September, and October.
- Some months with elevated storm activity also showed elevated property crime, but the pattern was inconsistent across all months.
- The moderate positive correlation was not statistically significant at $\alpha = 0.05$.
- The available 2024 data do not provide sufficient evidence to conclude that higher storm-event frequency was reliably associated with increased property crime.

The key analytical takeaway is the distinction between visual pattern recognition and statistical evidence. A relationship can appear plausible in a monthly chart while still failing to meet a statistical significance threshold. The results should also not be interpreted as evidence that storms cause changes in crime.

Limitations

- Only one calendar year of crime and storm data was available, limiting the ability to distinguish recurring patterns from one-year variation.
- Miami crime activity was compared with statewide Florida storm-event frequency rather than a directly matched city-level weather series.

- Property-crime classification required analyst judgment because the dataset did not provide a predefined property-crime category.
- Pearson correlation measures linear association and does not establish causation.
- Additional geographic and temporal detail would be needed before using the relationship for predictive decision-making.

Future Analysis

A stronger follow-up study could expand the project in several directions:

- Incorporate multiple years of crime and storm-event data to identify recurring seasonal patterns.
- Isolate hurricanes, tropical storms, and other severe-weather categories rather than treating all storm events equally.
- Use geographic or GIS-based matching to connect storm events more precisely with crime locations.
- Compare crime activity before, during, and after major storm events.
- Evaluate storm severity, seasonality, geography, and historical crime patterns within a multivariable or predictive model.

The original report also noted that severe weather can create conditions different from other storm types and that recurring seasonal patterns may be useful in predictive crime models (National Oceanic and Atmospheric Administration, n.d.; Sánchez-Delgado & Delgado, 2023).

Data Availability

The original crime and storm-event datasets were supplied through university coursework and are not redistributed in the public portfolio repository because no explicit redistribution license was provided. The portfolio repository instead presents the cleaned notebook, analytical workflow, derived monthly results, visualizations, and interpretation.

References

National Oceanic and Atmospheric Administration. (n.d.). Hurricane hazards.

<https://www.weather.gov/wrn/hurricane-hazards>

Sánchez-Delgado, H., & Delgado, R. (2023). The effect of seasonality in predicting the level of crime: A spatial perspective. PLOS ONE, 18(5). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10231786/>

Southern New Hampshire University. (2026). stormEvents2024.csv [Data set]. uCertify DAT 375 course materials.

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